

End-to-End Data Analytics Platform

Microsoft Fabric | Public Data | Spain Indicators

Formal definition of the five public data sources, their technical profiles, access patterns, and quality requirements for the end-to-end data analytics platform built on Microsoft Fabric.

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6 years experience	Lakehouse Warehouse	Duration: 1-2 months
Microsoft Stack	PySpark Power BI	Version 1.0

1. Executive Summary

This document formalizes the requirements and data source specifications for Phase 1 of the End-to-End Data Analytics Platform project. It serves as the definitive reference for all subsequent phases: architecture design, functional and technical documentation, development in Microsoft Fabric, and GitHub publication.

Five public data sources have been selected from Spanish government institutions, covering diverse domains (**energy, consumer prices, industrial production, public construction tenders, and tax revenue**) across five distinct ingestion formats (**REST API, CSV, CSV loaded into Azure SQL Database, legacy XLS, and complex XLSX**). Each source was chosen not only for its thematic relevance but also for the specific technical challenges it presents, ensuring the portfolio demonstrates real-world data engineering capabilities.

The geographic scope is limited to three entities: **Catalonia, Madrid, and National Total**, providing a consistent cross-source analytical dimension. The temporal scope is variable per source, using the maximum available historical depth each source permits.

2. Project Scope Summary

Functional Scope

- Ingest 5 public data sources from 4 Spanish government institutions.
- Implement Bronze-Silver-Gold medallion architecture in Microsoft Fabric Lakehouse.
- Transform and cleanse data using PySpark Notebooks and Dataflows Gen2.
- Model a star schema in Fabric Warehouse with fact and dimension tables.
- Deliver a Power BI report with Direct Lake connectivity.
- Orchestrate the complete pipeline using Data Pipelines.

Geographic Scope

Entity	Type	Scope Rationale
Catalonia	Autonomous Community	Author's current location (Barcelona). Regional economic relevance.
Madrid	Autonomous Community	Spain's economic and administrative capital. Largest tax revenue contributor.
National Total	Aggregated	Baseline for comparison and national-level analysis.

Temporal Scope

Each source uses the maximum available historical depth. This decision maximizes the analytical value of the dataset while demonstrating the ability to handle varying temporal granularities across sources. The specific availability window for each source is documented in the Source Catalog below.

3. Source Catalog

Each source is documented with a technical profile covering its origin, format, access pattern, structure, known challenges, and the rationale for its selection. Sources are ordered by ingestion complexity.

CSV

Source 1: Consumer Price Index (IPC)

Instituto Nacional de Estadística (INE)

Official Name	Índice de Precios de Consumo (IPC) — Base 2025
Institution	Instituto Nacional de Estadística (INE)
Format	CSV (semicolon-delimited, ISO-8859-15 encoding)
Access Method	Direct HTTP download (stable URL, no authentication)
URL	https://www.ine.es/jaxiT3/files/t/es/csv_bdsc/76136.csv?nocab=1
Update Frequency	Monthly
Granularity	Autonomous Community × ECOICOP Group × Data Type × Month
Temporal Depth	Full historical series available (2002–present)
Volume Estimate	~50,000+ rows (full dataset)

Technical Challenges

- Non-UTF-8 encoding (ISO-8859-15) requires explicit encoding handling during ingestion.
- Semicolon delimiter (;) instead of standard comma, requiring parser configuration.
- Period format 2026M02 needs parsing into proper date types (year + month extraction).
- Filtering by geographic scope (Nacional, Cataluña, Madrid) from the Autonomous Community column.

Selection Rationale

Clean, well-structured CSV that serves as the baseline ingestion pattern. Demonstrates standard CSV handling with encoding and delimiter specifics. The IPC is Spain's most widely referenced economic indicator, adding immediate thematic credibility to the portfolio.

Fabric Component

Data Pipeline (Copy Activity) for ingestion into Bronze layer. **Dataflow Gen2** for Silver layer transformation (light ETL: filtering, type casting, column renaming). This is an ideal candidate for Dataflow Gen2 because the transformations are simple and do not require Spark overhead.

CSV → SQL

Source 2: Industrial Production Index (IPI)

Instituto Nacional de Estadística (INE)

Official Name	Índice de Producción Industrial (IPI) — Base 2021
Institution	Instituto Nacional de Estadística (INE)
Native Format	CSV (semicolon-delimited, ISO-8859-15 encoding)
Ingestion Format	Azure SQL Database (CSV pre-loaded as staging table)
CSV URL	https://www.ine.es/jaxiT3/files/t/es/csv_bdsc/60272.csv?nocab=1
Azure SQL Tier	Free tier (32 GB storage, 100K vCore-seconds/month)
Update Frequency	Monthly
Granularity	Autonomous Community × Economic Destination × Index Type × Month
Temporal Depth	Full historical series available (2002–present)
Volume Estimate	~40,000+ rows (full dataset)

Technical Challenges

- Requires provisioning and configuring an Azure SQL Database as the source system.
- CSV must be pre-loaded into Azure SQL (manual or scripted bulk insert) to simulate a transactional source.
- Cloud-to-cloud connectivity from Microsoft Fabric to Azure SQL via Data Pipeline.
- Same encoding and delimiter challenges as Source 1 during the initial CSV-to-SQL load.

Selection Rationale

This source serves a dual purpose: it demonstrates **SQL Server ingestion** (a common enterprise pattern) and **cloud-to-cloud connectivity**. By pre-loading the INE CSV into Azure SQL Database, the project simulates a real-world scenario where Fabric ingests from an existing transactional database. The IPI complements the IPC thematically, providing the industrial dimension to Spain's economic picture.

Fabric Component

Data Pipeline (Copy Activity) with Azure SQL Database connector for ingestion into Bronze layer. **Dataflow Gen2** for Silver layer transformation. The Azure SQL setup is documented separately to ensure reproducibility.

REST API

Source 3: Electricity Demand Evolution

Red Eléctrica de España (REE) — Redeia Group

Official Name	Evolución de la Demanda Eléctrica
Institution	Red Eléctrica de España (REE) — Redeia Group
Format	REST API (JSON, JSONAPI standard)
Access Method	Public REST API, no authentication required
Base URL	https://apidatos.ree.es
Endpoint	/es/datos/demanda/evolucion
Key Parameters	start_date, end_date, time_trunc, geo_trunc, geo_limit, geo_ids
Update Frequency	Daily (real-time data available)
Granularity	Configurable: hour / day / month / year. Filterable by CCAA.
Temporal Depth	2014–present (varies by indicator)
Rate Limiting	No documented rate limit, but responsible usage recommended.

API Request Pattern

Example request for monthly demand evolution in Catalonia (2023–2025):

```
GET /es/datos/demanda/evolucion?start_date=2023-01-01T00:00&end_date=2025-12-31T23:59&time_trunc=month&geo_trunc=electric_system&geo_limit=ccaa&geo_ids=9
```

Region	geo_limit	geo_ids
National (Peninsular)	peninsular	8741
Catalonia	ccaa	9
Madrid	ccaa	13

Technical Challenges

- Nested JSON response structure: data → included[] → attributes → values[] requires recursive parsing.
- API has maximum date range constraints; large historical requests may need to be paginated by year.
- Multiple indicators returned per request (demand types) need flattening into tabular format.
- Three separate API calls required (one per geographic entity) to build the complete dataset.

Selection Rationale

The only REST API source in the project. Demonstrates API-based ingestion via PySpark Notebook, including HTTP request handling, JSON parsing, pagination logic, and error management. The energy domain diversifies the portfolio beyond purely economic indicators and connects to Spain's ongoing energy transition narrative.

Fabric Component

Notebook (PySpark/Python) for ingestion into Bronze layer. The API's nested JSON structure and pagination requirements justify using a Notebook over a Data Pipeline Copy Activity. **Notebook** (PySpark) also handles Silver layer transformation (flattening, type enforcement, deduplication).

Source 4: Official Public Construction Tenders

XLS

Ministerio de Transportes y Movilidad Sostenible

Official Name	Licitación Oficial en Construcción — Total Construcción
Institution	Ministerio de Transportes y Movilidad Sostenible
Format	XLS (Excel 97-2003 binary format, .xls extension)
Access Method	HTTP download from the Ministry's Statistical Bulletin
Update Frequency	Monthly
Temporal Depth	Full historical series (typically 2000–present)

File Inventory

File	Content	Geographic Scope
01401400.XLS	State & Social Security — Total Construction	National + Catalonia
01401600.XLS	State & Social Security — Total Construction	Madrid
01402600.XLS	Territorial Entities — Total Construction	National + Catalonia
01402800.XLS	Territorial Entities — Total Construction	Madrid

Technical Challenges

- Legacy XLS binary format (Excel 97-2003) requires specific libraries (xlrd) for reading.
- Complex header structures with merged cells, multi-row headers, and embedded subtotals.
- Four separate files must be consolidated into a single unified dataset.
- Geographic split across files: Madrid in separate files from Nacional+Cataluña.
- Data cleaning required: remove subtotal rows, handle empty cells, normalize column names.
- Contracting agent dimension (State vs. Territorial Entities) adds a categorical split.

Selection Rationale

This is the **highest-complexity ingestion source** in the project and the one that best demonstrates real-world data engineering skills. Legacy Excel files with merged cells, inconsistent layouts, and multi-file consolidation are exactly the kind of challenge that separates tutorial-level portfolios from professional ones. The construction domain adds a third thematic axis to the project.

Fabric Component

Notebook (PySpark/Python) for both Bronze ingestion and Silver transformation. The file complexity (merged cells, multi-row headers, subtotals, multi-file consolidation) makes this unsuitable for Dataflow Gen2 or simple Copy Activity. A Notebook provides the programmatic control needed for reliable parsing.

XLSX

Source 5: Tax Revenue by Regional Offices

Agencia Estatal de Administración Tributaria (AEAT)

Official Name	Ingresos por Delegaciones y Servicios Centrales
Institution	Agencia Estatal de Administración Tributaria (AEAT)
Format	XLSX (modern Excel, complex multi-sheet workbook, ~6 MB)
Access Method	HTTP download from AEAT open data portal
Target Sheet	datos_delegaciones (last sheet in the workbook)
Update Frequency	Monthly
Granularity	Delegation × Tax Concept × Period
Temporal Depth	Full historical series available
Volume Estimate	Large workbook with multiple sheets; target sheet contains structured data.

Technical Challenges

- Large XLSX workbook (~6 MB) with multiple sheets; only datos_delegaciones is in scope.
- Complex internal structure: data may be in crosstab/pivot format requiring unpivot operations.
- Delegation-to-Autonomous Community mapping required (56 delegations → 17 CCAA + 2 cities).
- Multi-level column headers that need programmatic parsing.
- Tax concept hierarchy needs careful handling for meaningful aggregation.

Selection Rationale

Demonstrates handling of complex modern Excel files: multi-sheet workbooks, selective sheet extraction, crosstab-to-tabular transformation, and reference data mapping (delegation → CCAA). Tax revenue adds the fiscal dimension to the project, and AEAT is one of Spain's most authoritative public data sources, adding institutional credibility.

Fabric Component

Notebook (PySpark/Python) for Bronze ingestion (selective sheet reading) and Silver transformation (unpivot, delegation mapping, type enforcement). The complexity of the Excel structure justifies Notebook usage over Dataflow Gen2.

4. Source Comparison Matrix

The following matrix provides a consolidated view of all five sources, highlighting the deliberate variety in formats, ingestion patterns, and Fabric components used.

#	Source	Format	Ingestion	Transformation	Complexity
1	IPC (INE)	CSV	Pipeline (Copy Activity)	Dataflow Gen2	Low
2	IPI (INE)	CSV → Azure SQL	Pipeline (SQL Connector)	Dataflow Gen2	Medium
3	Energy Demand (REE)	REST API (JSON)	Notebook (Python)	Notebook (PySpark)	Medium- High
4	Construction Tenders (MITMA)	XLS (Legacy)	Notebook (Python)	Notebook (PySpark)	High
5	Tax Revenue (AEAT)	XLSX (Complex)	Notebook (Python)	Notebook (PySpark)	High

Key Design Principle: Each Fabric component is used where it naturally fits. Dataflows Gen2 handle simple, low-complexity transformations (Sources 1–2) where Spark overhead is unnecessary. Notebooks handle complex ingestion and transformation (Sources 3–5) where programmatic control is required. Data Pipelines orchestrate the entire flow. Nothing is forced to demonstrate tool usage; each choice is justified by the nature of the data and the transformation required.

5. Target Model Preview

The final analytical layer will be implemented as a **star schema** in the Fabric Warehouse, optimized for Power BI Direct Lake connectivity. The following is a preliminary outline; detailed specifications will be delivered in Phase 3 (Functional) and Phase 4 (Technical).

Fact Table

Table	Grain	Key Measures
fact_indicators	One row per indicator × region × period	indicator_value, percentage_change, absolute_change

Dimension Tables

Table	Description	Key Attributes
dim_indicator	Indicator metadata and classification	indicator_name, domain, source, unit_of_measure, frequency
dim_region	Geographic entities (CCAA)	region_name, region_code, region_type, country
dim_calendar	Time dimension	date, year, month, month_name, quarter, is_current_year

This model consolidates all five sources into a single analytical fact table, enabling cross-domain analysis (e.g., correlating energy demand trends with industrial production or construction activity). The dimension tables provide consistent filtering and slicing across all indicators.

6. Key Decisions

Variable temporal scope per source

Maximizes analytical depth without artificially limiting sources that have longer histories. Each source's available window is documented in its technical profile.

Azure SQL Database (free tier) for SQL source

Cloud-to-cloud ingestion is the modern standard. Azure SQL free tier provides 32 GB and 100K vCore-seconds/month at no cost, and Fabric connects natively via Data Pipeline. This avoids the narrative inconsistency of using on-premise sources in a cloud-native project.

Single sheet (datos_delegaciones) from AEAT workbook

Scoping to one sheet prevents over-engineering while still demonstrating complex XLSX handling. The selected sheet contains the structured data needed for the dimensional model.

Dataflow Gen2 for simple sources, Notebooks for complex sources

This is not an arbitrary split. Sources 1–2 (CSV, SQL) have clean structures that map directly to Dataflow Gen2's visual transformation interface. Sources 3–5 require programmatic control (API pagination, merged cells, unpivot) that only Notebooks provide.

7. Out of Scope

- Real-time or scheduled refresh pipelines (this is a one-time load for portfolio demonstration).
- The full 80+ source dataset from the original client project.
- Data quality frameworks or automated testing suites.
- Row-Level Security (RLS), workspace roles, or multi-tenant configurations.
- Advanced analytics, machine learning, or predictive modeling.
- Incremental loading patterns (full refresh is sufficient for portfolio scope).
- Autonomous Communities beyond the defined geographic scope (Catalonia, Madrid, National).

8. Next Steps

With the requirements and source catalog formalized, the project proceeds to **Phase 2: Architecture Design**, which will deliver:

- High-level architecture diagram of the complete Fabric solution.
- Detailed data flow diagram showing Bronze → Silver → Gold movement per source.
- Component mapping: which Fabric item handles each step of the pipeline.
- Naming conventions for all Fabric items (Lakehouse tables, Warehouse objects, Pipelines, Notebooks).

0	Context & Strategy Deliverable: Project Brief (PDF)	COMPLETE
1	Requirements Deliverable: Requirements Document + Source Catalog	COMPLETE
2	Architecture Deliverable: Architecture Diagram + Design Document	NEXT
3	Functional Spec Deliverable: Functional Document (PDF)	PENDING
4	Technical Spec Deliverable: Technical Document (PDF)	PENDING
5	Development Deliverable: Working Solution + Code	PENDING
6	Publication Deliverable: GitHub Repository + README	PENDING

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